

MIAE 221 · Rapid Review Sheet · Part 7

Dislocations & Strengthening Mechanisms

1. Slip Systems & Crystal Architecture

- **Rule:** Plastic deformation occurs by dislocation glide on close-packed planes along close-packed directions.
- **Slip System Comparison:**

Structure	Slip Planes	Slip Directions	Total Systems	Ductility Character
FCC	111 (4)	$\langle 110 \rangle$ (3)	12	Highly ductile at all temperatures (satisfies Von Mises ≥ 5).
BCC	110, 112, 123	$\langle 111 \rangle$ (2)	48	Strong; pronounced Ductile-to-Brittle Transition (DBTT) at low T .
HCP	0001 basal (1)	$\langle 11\bar{2}0 \rangle$ (3)	3	Brittle at room temperature (< 5 systems; twinning required).

2. Schmid's Law & Single Crystal Yielding

- **Resolved Shear Stress:** $\tau_R = \sigma \cos \phi \cos \lambda = \sigma \cdot m$
 - ϕ = Angle between tensile axis and slip plane normal.
 - λ = Angle between tensile axis and slip direction.
 - $m = \cos \phi \cos \lambda$ = **Schmid Factor** ($\phi + \lambda \geq 90^\circ$).
- **Critical Resolved Shear Stress (τ_{crss}):** Minimum shear stress to initiate slip:

$$\sigma_y = \frac{\tau_{\text{crss}}}{\cos \phi \cos \lambda} = \frac{\tau_{\text{crss}}}{m}$$

- **Maximum Schmid Factor:** At $\phi = \lambda = 45^\circ \implies m_{\text{max}} = 0.50 \implies \sigma_{y,\text{min}} = 2, \tau_{\text{crss}}$.

3. The 4 Metallurgical Strengthening Mechanisms

- **Core Principle:** Impede dislocation motion to increase yield strength!

1. Grain Size Reduction (Hall-Petch):

$$\sigma_y = \sigma_0 + k_y \cdot d^{-1/2}$$

- Dislocation pile-ups at grain boundaries. ONLY mechanism that increases strength **AND** toughness!

2. Solid Solution Strengthening:

- Solute atom size misfit (Δr) creates compressive/tensile lattice strain fields that pin dislocation cores.

3. Strain Hardening / Cold Work:

- Dislocation density explodes: $\rho_d \approx 10^6 \rightarrow 10^{10} \text{ cm}^{-2}$. Dislocation tangles mutually obstruct slip ($\tau_y \propto Gb\sqrt{\rho_d}$).

4. Precipitation Hardening:

- Nanoscale particles impede dislocations via cutting or **Orowan looping** ($\tau \approx Gb/L$).

4. Annealing Stages of Cold-Worked Metals

Stage	Temperature Range	Microstructure	Mechanical Property Evolution
1. Recovery	$T < T_R (\approx 0.1 - 0.3, T_m)$	Dislocation annihilation & polygonization. Grains unchanged.	Residual stresses relieve; conductivity restored; strength unchanged.
2. Recrystallization	$T_R \approx 0.3 - 0.4, T_m$	Nucleation of new strain-free equiaxed grains consuming cold-worked grains.	σ_y and UTS drop sharply; Ductility fully restored!
3. Grain Growth	$T \gg T_R$	Large grains coarsen ($d^n - d_0^n = Kt$) to reduce boundary energy.	Yield strength decreases slightly per Hall-Petch.